CS5998 Capstone Project Definition Document

Predictive Maintenance: Predicting Remaining Useful Life (RUL) in Industrial Turbofan Engines

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1. Problem Statement

1.1. Operational Challenge

Unplanned mechanical failures in industrial turbofan engines can result in substantial financial losses, flight disruptions, and serious safety risks. Traditional maintenance strategies generally rely on either reactive maintenance, where repairs are carried out after a failure occurs, or scheduled maintenance, where components are overhauled at predefined intervals. Although scheduled overhauls help reduce the risk of unexpected failures, they can lead to the premature replacement of components that are still in good working condition. In contrast, reactive maintenance may allow a failure to develop further, potentially causing additional damage to surrounding equipment and increasing repair costs and operational downtime.

1.2. Objective & Capstone Bounded Framework

This project aims to develop an end-to-end predictive maintenance system that uses multi-sensor telemetry data to estimate the Remaining Useful Life (RUL) of turbofan engines. By analyzing changes in sensor measurements over time, the system seeks to identify patterns associated with component degradation and provide engineering teams with early warnings before the engine reaches a critical condition. In line with the capstone project guidelines, the project is defined within the following scope:

Problem Type: Continuous time-series regression for cycle-level RUL forecasting. Operational maintenance alerts will be generated by applying predefined thresholds to the continuous RUL predictions.

Data Type: The system uses multivariate industrial sensor telemetry collected over time, together with multiple operational settings that describe the conditions under which each engine operates.

Technique Category: The project compares classical tree-based machine learning methods with deep learning models designed to capture temporal patterns and dependencies within sequential data.

System Context: The proposed solution includes batch and streaming data-processing pipelines, a containerized REST-based scoring API for serving model predictions, and an operational monitoring dashboard for viewing prediction results and system status.

2. Data Source(s)

2.1. Benchmark Selection: NASA CMAPSS

The proposed system uses the publicly available NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset. CMAPSS is a well-established benchmark dataset for research in equipment health monitoring and Remaining Useful Life (RUL) prediction.

Primary Subset: The FD001 subset consists of 100 training trajectories containing 20,631 run-to-failure cycles and 100 test trajectories covering 13,096 cycles, along with the corresponding ground-truth validation vectors. This subset represents a single operational condition and focuses on the degradation of the High-Pressure Compressor (HPC).

Secondary Subset: The FD002 subset contains 260 training trajectories and 259 test trajectories. It is used to assess the stability of the models under six different operational flight conditions.

Feature Topology: Each data record contains 26 raw columns, including the Unit ID, operational cycle number, three operational settings (Altitude, Mach number, and Throttle Resolver Angle), and 21 sensor measurements covering thermal, pressure, and rotational characteristics.

2.2. Data Preprocessing Strategy

RUL Target Labelling: The pipeline uses a piecewise-linear RUL target with a maximum value of 125 cycles. This reflects the assumption that mechanical degradation is relatively limited during the early stages of operation. Capping the RUL value also helps prevent the models from learning unrealistic degradation patterns from engines that are still in a healthy operating condition.

Feature Engineering: Sensors with zero variance are first removed to eliminate channels that provide no useful information. For the FD001 subset, this includes sensors 1, 5, 6, 10, 16, 18, and 19. The remaining sensor data is then used to calculate rolling temporal statistics, including the mean, standard deviation, minimum, and maximum, over a historical window of 30 operating cycles.

Leakage Prevention: Z-score and Min-Max scaling are fitted exclusively using the training engine units. The same fitted transformations are subsequently applied to the validation and test units. This approach ensures that information from the validation and test data does not influence the training process and helps prevent data leakage.

3. Intended Methods (Linked to Program Modules)

3.1. Module Linkages

Data Mining: Exploratory analysis of sensor relationships, feature selection, sliding-window transformations, and dimensionality reduction will be used to develop a meaningful health index.

Machine Learning and Deep Learning: Tree-based regression models will be developed as baseline methods, followed by the development and optimization of temporal 1D-CNN and Bidirectional LSTM models.

Cloud Computing and Distributed Systems: The system will use Docker containers to deploy inference services and provide automated batch-scoring endpoints.

3.2. Comparative Modeling Architecture

To meet the required milestone objectives, the project compares a conventional machine-learning approach with a deep sequential architecture that is designed to capture temporal patterns in engine degradation.

Baseline Architecture: XGBoost and Random Forest regression models will be trained using tabular features generated from rolling windows of 30 operating cycles.

Advanced Architecture: A hybrid 1D-CNN and BiLSTM sequential model will be developed to learn degradation patterns directly from the sensor sequences. The 1D-CNN layers will capture relationships

between the different sensor channels, while the stacked bidirectional LSTM layers will learn longer-term temporal patterns associated with engine degradation.

3.3. Evaluation Metrics

The models are evaluated on the test trajectories using two complementary metrics:

Root Mean Squared Error (RMSE): Measures how closely the predicted RUL values correspond to the actual RUL values across the operating cycles.

NASA Scoring Metric: Uses an asymmetric exponential penalty to reflect the practical requirements of industrial maintenance. Predictions that overestimate the actual RUL, which may delay necessary maintenance and increase the risk of failure, receive a substantially higher penalty than predictions that underestimate the RUL.

4. Expected Outputs & Milestone Deliverables

Deliverables map directly to the project milestone framework across the 12-week project cycle:

Milestone	Timeline	Key Deliverables
Milestone 1 (25%)	Week 4	Project Definition Document (scope lock), public GitHub repository established with initial commit history, raw data acquisition pipeline.
Milestone 2 (25%)	Week 8	EDA notebook, completed sliding-window pipeline, trained baseline model (XGBoost), trained advanced model (1D-CNN+BiLSTM), architecture diagram, and 6-minute recorded technical demo.
Milestone 3 (50%)	Week 12	Final Technical Report (25-30 pages), Reproducibility Checklist (Appendix A verified), Dockerized FastAPI scoring service, and Streamlit operational fleet monitoring UI.

5. Key Risks and Assumptions

Category	Assumed Condition	Risk / Failure Mode	Mitigation Strategy
Data Quality	Complete telemetry streams without missing cycles.	Sensor noise or missing values corrupt sliding sequence arrays.	Implement robust forward-fill and channel-median imputation during ingestion.
Model Drift	Linear degradation holds past knee-point (RUL=125).	Improper threshold penalizes healthy early-cycle engines.	Perform cross-validation grid search on wear thresholds (RUL in [110, 140] cycles).
Asymmetric Loss	Standard MSE/RMSE reflects real maintenance costs.	Symmetric metrics reward models that underestimate catastrophic late failure.	Evaluate and checkpoint models directly against the NASA asymmetric exponential penalty metric.